OSCILLATIONS, WAVES AND OPTICS

PHYSICS FOR CHEMISTRY STUDENTS

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4.1 ELECTROMAGNETIC WAVES

Maxwell's Eqs:

$$\boldsymbol{\nabla} \times \boldsymbol{E}(\boldsymbol{r},t) = -\frac{\partial \boldsymbol{B}(\boldsymbol{r},t)}{\partial t}$$

$$\boldsymbol{\nabla} \times \boldsymbol{H}(\boldsymbol{r},t) = \frac{\partial \boldsymbol{D}(\boldsymbol{r},t)}{\partial t} + \boldsymbol{j}(\boldsymbol{r},t)$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{D}(\boldsymbol{r},t) = \rho(r,t)$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B}(\boldsymbol{r},t) = 0$$

$$\boldsymbol{H}(r,t) = {\mu_0}^{-1}\boldsymbol{B}(\boldsymbol{r},t) - \boldsymbol{M}(\boldsymbol{r},t)$$

$$\boldsymbol{D}(r,t) = \varepsilon_0 \boldsymbol{E}(\boldsymbol{r},t) + \boldsymbol{P}(\boldsymbol{r},t) = \varepsilon \boldsymbol{E}(\boldsymbol{r},t)$$

4.1 ELECTROMAGNETIC WAVES

Divergence | div | $\boldsymbol{\nabla} \cdot \boldsymbol{A} = \left(\frac{\partial}{\partial x}\boldsymbol{i} + \frac{\partial}{\partial y}\boldsymbol{j} + \frac{\partial}{\partial z}\boldsymbol{k}\right) \cdot (A_x, A_y, A_z)$

Curl | curl or rot | $\boldsymbol{\nabla} \times \boldsymbol{A} = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$

Gradient | grad | $\nabla f = \frac{\partial f}{\partial x}\boldsymbol{i} + \frac{\partial f}{\partial y}\boldsymbol{j} + \frac{\partial f}{\partial z}\boldsymbol{k}$

$$\nabla \times (\nabla \times \boldsymbol{A}) = \nabla(\nabla \cdot \boldsymbol{A}) - \nabla^2 \boldsymbol{A}$$

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) = 0 \qquad \boldsymbol{\nabla} \times (\nabla f) = 0$$

https://en.wikipedia.org/wiki/Divergence

4.1 ELECTROMAGNETIC WAVES

In free space:

$$\boldsymbol{H}(r,t) = \mu_0{}^{-1}\boldsymbol{B}(\boldsymbol{r},t) \qquad \boldsymbol{D}(r,t) = \varepsilon_0\boldsymbol{E}(\boldsymbol{r},t)$$

$$\boldsymbol{\nabla} \times \boldsymbol{E}(\boldsymbol{r},t) = -\frac{\partial \boldsymbol{B}(\boldsymbol{r},t)}{\partial t}$$

$$\boldsymbol{\nabla} \times \boldsymbol{H}(\boldsymbol{r},t) = \frac{\partial \boldsymbol{D}(\boldsymbol{r},t)}{\partial t} + \boldsymbol{j}(\boldsymbol{r},t)$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{D}(\boldsymbol{r},t) = \rho(r,t)$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B}(\boldsymbol{r},t) = 0$$

$$\boldsymbol{\nabla} \times \boldsymbol{E}(\boldsymbol{r},t) = -\frac{\partial \boldsymbol{B}(\boldsymbol{r},t)}{\partial t}$$

$$\boldsymbol{\nabla} \times \boldsymbol{B}(\boldsymbol{r},t) = \varepsilon_0\mu_0\frac{\partial \boldsymbol{E}(\boldsymbol{r},t)}{\partial t}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{E}(\boldsymbol{r},t) = 0$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B}(\boldsymbol{r},t) = 0$$

$$\nabla \times (\nabla \times \boldsymbol{A}) = \nabla(\nabla \cdot \boldsymbol{A}) - \nabla^2\boldsymbol{A}$$

$$\nabla \times \nabla \times \boldsymbol{B}(\boldsymbol{r},t) = \nabla(\nabla \cdot \boldsymbol{B}) - \nabla^2\boldsymbol{B} = \varepsilon_0\mu_0\frac{\partial}{\partial t}\nabla \times \boldsymbol{E}(\boldsymbol{r},t)$$

$$-\nabla^2\boldsymbol{B}(\boldsymbol{r},t) = -\varepsilon_0\mu_0\frac{\partial^2}{\partial t^2}\boldsymbol{B}(\boldsymbol{r},t)$$

4.1 ELECTROMAGNETIC WAVES

In free space:

$$\boldsymbol{H}(r,t) = {\mu_0}^{-1}\boldsymbol{B}(\boldsymbol{r},t) \qquad \boldsymbol{D}(r,t) = \varepsilon_0\boldsymbol{E}(\boldsymbol{r},t)$$

$$\boldsymbol{\nabla} \times \boldsymbol{E}(\boldsymbol{r},t) = -\frac{\partial \boldsymbol{B}(\boldsymbol{r},t)}{\partial t}$$

$$\boldsymbol{\nabla} \times \boldsymbol{H}(\boldsymbol{r},t) = \frac{\partial \boldsymbol{D}(\boldsymbol{r},t)}{\partial t} + \boldsymbol{j}(\boldsymbol{r},t)$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{D}(\boldsymbol{r},t) = \rho(r,t)$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B}(\boldsymbol{r},t) = 0$$

$$\boldsymbol{\nabla} \times \boldsymbol{E}(\boldsymbol{r},t) = -\frac{\partial \boldsymbol{B}(\boldsymbol{r},t)}{\partial t}$$

$$\boldsymbol{\nabla} \times \boldsymbol{B}(\boldsymbol{r},t) = \varepsilon_0\mu_0\frac{\partial \boldsymbol{E}(\boldsymbol{r},t)}{\partial t}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{E}(\boldsymbol{r},t) = 0$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B}(\boldsymbol{r},t) = 0$$

$$\nabla \times (\nabla \times \boldsymbol{A}) = \nabla(\nabla \cdot \boldsymbol{A}) - \nabla^2\boldsymbol{A}$$

$$\nabla \times \nabla \times \boldsymbol{E}(\boldsymbol{r},t) = \nabla(\nabla \cdot \boldsymbol{E}) - \nabla^2\boldsymbol{E} = -\frac{\partial}{\partial t}\nabla \times \boldsymbol{B}(\boldsymbol{r},t)$$

$$-\nabla^2\boldsymbol{E}(\boldsymbol{r},t) = -\varepsilon_0\mu_0\frac{\partial^2}{\partial t^2}\boldsymbol{E}(\boldsymbol{r},t)$$

4.1 ELECTROMAGNETIC WAVES

In transparent glass:

$$\boldsymbol{H}(r,t) = \mu^{-1}\boldsymbol{B}(\boldsymbol{r},t) \qquad \boldsymbol{D}(r,t) = \varepsilon\boldsymbol{E}(\boldsymbol{r},t)$$

$$\mu = \mu_r\mu_0 \qquad \varepsilon = \varepsilon_r\varepsilon_0$$

$$\boldsymbol{\nabla}\times\boldsymbol{E}(\boldsymbol{r},t) = -\frac{\partial\boldsymbol{B}(\boldsymbol{r},t)}{\partial t}$$

$$\boldsymbol{\nabla}\times\boldsymbol{H}(\boldsymbol{r},t) = \frac{\partial\boldsymbol{D}(\boldsymbol{r},t)}{\partial t} + \boldsymbol{j}(\boldsymbol{r},t)$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{D}(\boldsymbol{r},t) = \rho(r,t)$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{B}(\boldsymbol{r},t) = 0$$

$$\boldsymbol{\nabla}\times\boldsymbol{E}(\boldsymbol{r},t) = -\frac{\partial\boldsymbol{B}(\boldsymbol{r},t)}{\partial t}$$

$$\boldsymbol{\nabla}\times\boldsymbol{B}(\boldsymbol{r},t) = \varepsilon\mu\frac{\partial\boldsymbol{E}(\boldsymbol{r},t)}{\partial t}$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{E}(\boldsymbol{r},t) = 0$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{B}(\boldsymbol{r},t) = 0$$

$$\nabla\times(\nabla\times\boldsymbol{A}) = \nabla(\nabla\cdot\boldsymbol{A}) - \nabla^2\boldsymbol{A}$$

$$\nabla\times\nabla\times\boldsymbol{E}(\boldsymbol{r},t) = \nabla(\nabla\cdot\boldsymbol{E}) - \nabla^2\boldsymbol{E} = -\frac{\partial}{\partial t}\nabla\times\boldsymbol{B}(\boldsymbol{r},t)$$

$$-\nabla^2\boldsymbol{E}(\boldsymbol{r},t) = -\varepsilon\mu\frac{\partial^2}{\partial t^2}\boldsymbol{E}(\boldsymbol{r},t)$$

4.1 ELECTROMAGNETIC WAVES

Wave equations

$$\frac{\partial^2}{\partial t^2}\boldsymbol{B}(r,t) = \frac{1}{\varepsilon_0\mu_0}\nabla^2\boldsymbol{B}(r,t)$$

$$\frac{\partial^2}{\partial t^2}\boldsymbol{E}(r,t) = \frac{1}{\varepsilon_0\mu_0}\nabla^2\boldsymbol{E}(r,t)$$

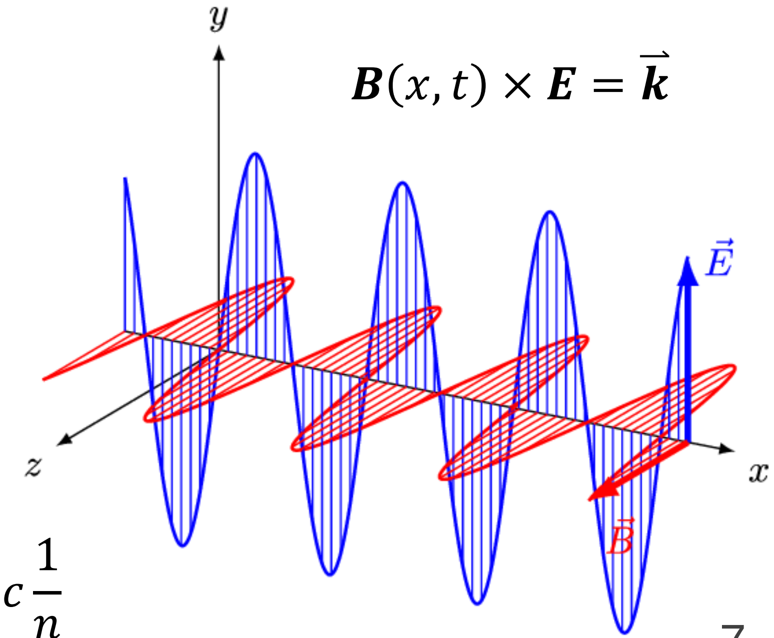
Solutions:

$$\boldsymbol{E}(r,t) = \boldsymbol{E_0}e^{ikr-i\omega t}$$

$$\boldsymbol{B}(r,t) = \boldsymbol{B_0}e^{ikr-i\omega t}$$

$$v_p = \frac{\omega}{k} = \sqrt{\frac{1}{\varepsilon_0\mu_0}}$$

$$\boldsymbol{B}(x,t) \times \boldsymbol{E} = \vec{\boldsymbol{k}}$$



Transverse wave

Free space

$$v_p = \sqrt{\frac{1}{\varepsilon_0\mu_0}} = c$$

Glass

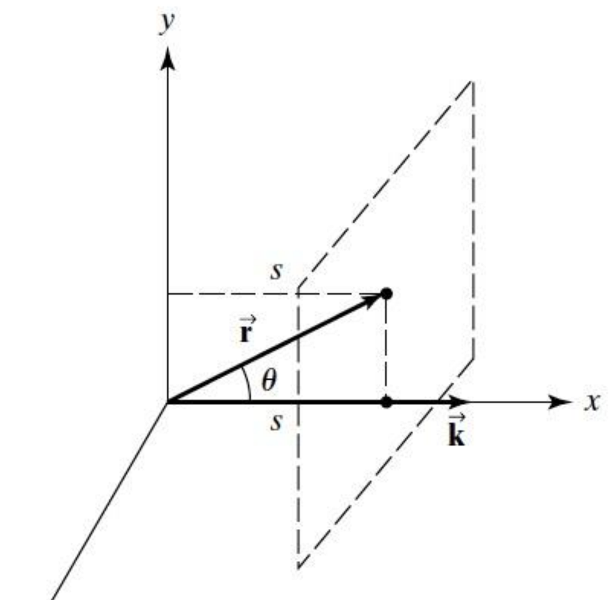
$$v_p = \sqrt{\frac{1}{\varepsilon_0\varepsilon_r\mu_0\mu_r}} = c\sqrt{\frac{1}{\varepsilon_r\mu_r}} = c\frac{1}{n}$$

4.1 ELECTROMAGNETIC WAVES

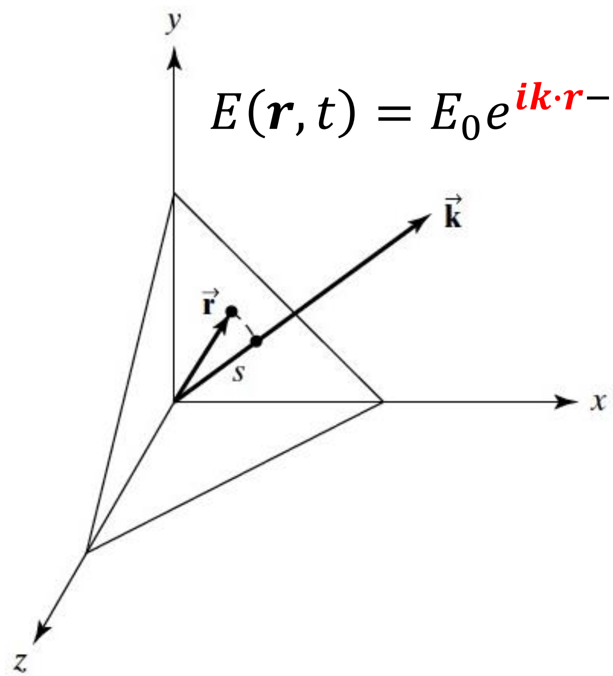
Plane waves

$$\boldsymbol{k}\cdot\boldsymbol{r} = xk_x + yk_y + zk_z = kr\cos\theta \equiv ks$$

$$E(\boldsymbol{r},t) = E_0 e^{\color{red}{i\boldsymbol{k}\cdot\boldsymbol{r}}-i\omega t}$$



$$E(\boldsymbol{r},t) = E_0 e^{ikx-i\omega t}$$



$$E(\boldsymbol{r},t) = E_0 e^{\color{red}{i\boldsymbol{k}\cdot\boldsymbol{r}}-i\omega t}$$

4.1 ELECTROMAGNETIC WAVES

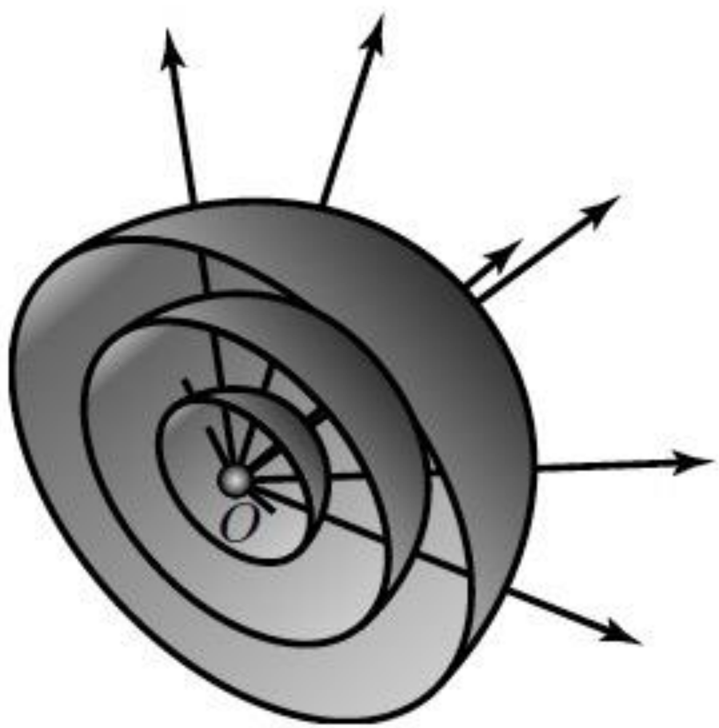
Spherical waves

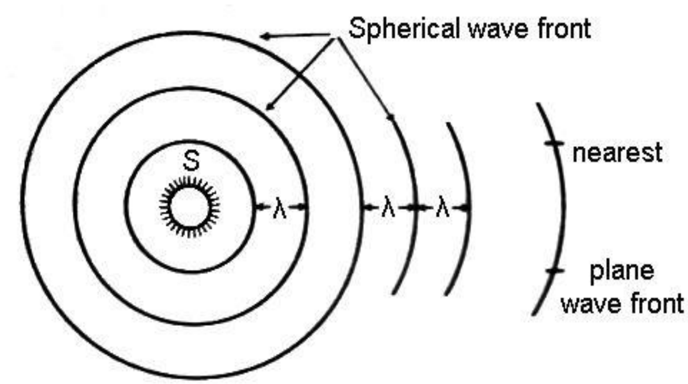
$$E(r,t) = \frac{E_0}{r} e^{ikr - i\omega t}$$

solution

$$\frac{\partial^2}{\partial t^2}\boldsymbol{E}(r,t) = \frac{1}{\varepsilon_0 \mu_0}\nabla^2 \boldsymbol{E}(r,t)$$

$$\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r} + \frac{1}{r^2 \sin^2\phi}\frac{\partial^2}{\partial \theta^2} + \frac{\cos\phi}{r^2 \sin\phi}\frac{\partial}{\partial \phi} + \frac{1}{r^2}\frac{\partial^2}{\partial \phi^2} .$$



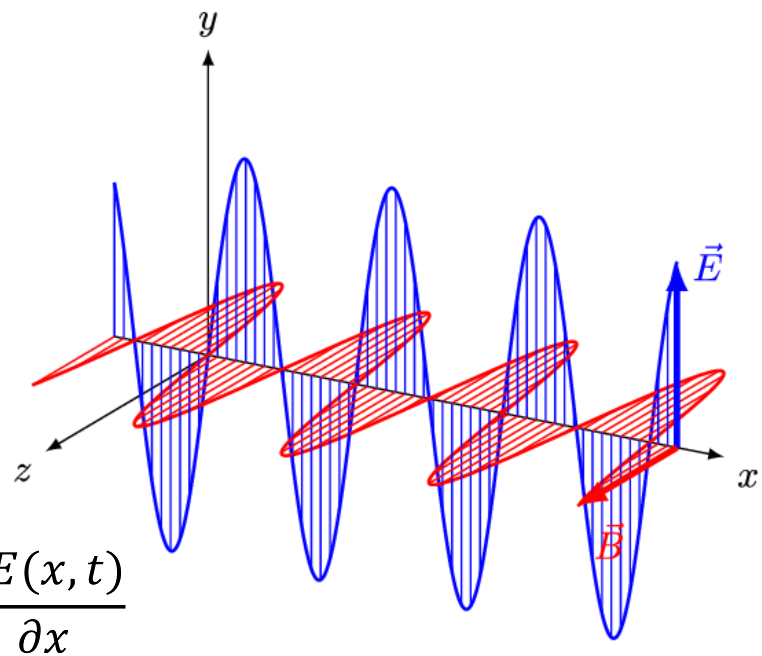


4.1 ELECTROMAGNETIC WAVES

$$\boldsymbol{E}(x,t) = \boldsymbol{E_0} e^{i\boldsymbol{kx} - i\omega t}$$

$$\boldsymbol{B}(x,t) = \boldsymbol{B_0} e^{i\boldsymbol{kx} - i\omega t}$$

$$\boldsymbol{\nabla} \times \boldsymbol{E}(x,t) = -\frac{\partial \boldsymbol{B}(x,t)}{\partial t}$$



$$\boldsymbol{\nabla} \times \boldsymbol{E} = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & E_y & 0 \end{vmatrix} = \widehat{\boldsymbol{k}} \frac{\partial E(x,t)}{\partial x}$$

$$-\frac{\partial \boldsymbol{B}(x,t)}{\partial t} = \widehat{\boldsymbol{k}} \frac{\partial E(x,t)}{\partial x}$$

4.1 ELECTROMAGNETIC WAVES

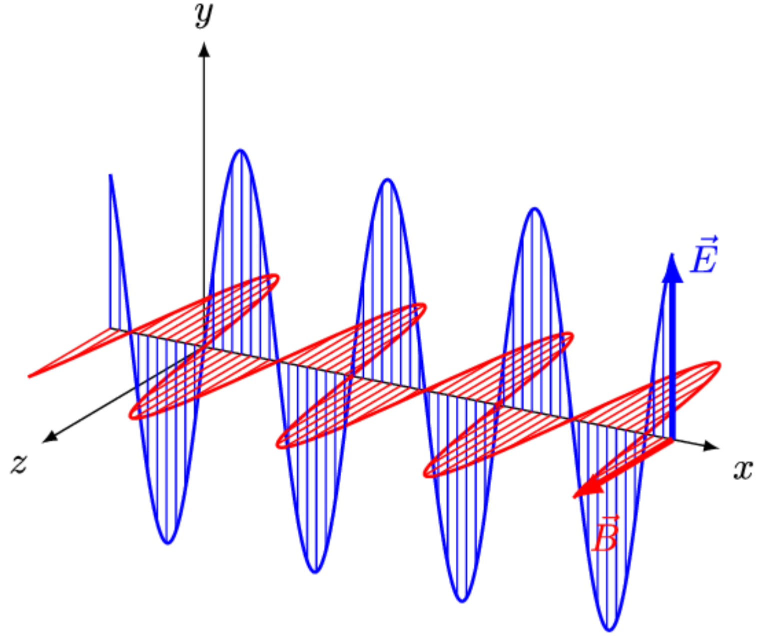
$$\boldsymbol{E}(x,t) = \boldsymbol{E_0} e^{i\boldsymbol{kx} - i\omega t}$$

$$\boldsymbol{B}(x,t) = \boldsymbol{B_0} e^{i\boldsymbol{kx} - i\omega t}$$

$$\boldsymbol{\nabla} \times \boldsymbol{E}(x,t) = -\frac{\partial \boldsymbol{B}(x,t)}{\partial t}$$

$$ikE(x,t) = i\omega B(x,t)$$

$$E(x,t) = cB(x,t)$$



Energy density

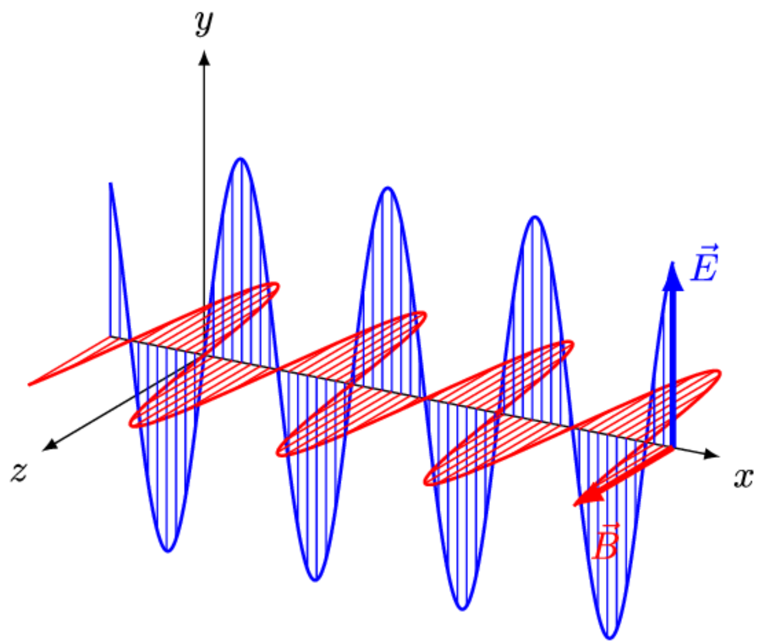
$$\epsilon_E = \frac{1}{2}\varepsilon_0 (E)^2 \qquad \epsilon_B = \frac{1}{2}\frac{1}{\mu_0}(B)^2 \qquad \sqrt{\frac{1}{\varepsilon_0 \mu_0}} = c \qquad \epsilon_E = \epsilon_B$$

4.1 ELECTROMAGNETIC WAVES

Energy density

$$\epsilon = \epsilon_E + \epsilon_B = \varepsilon_0 (E)^2$$

$$\langle \epsilon \rangle = \frac{1}{2} \varepsilon_0 (E)^2$$

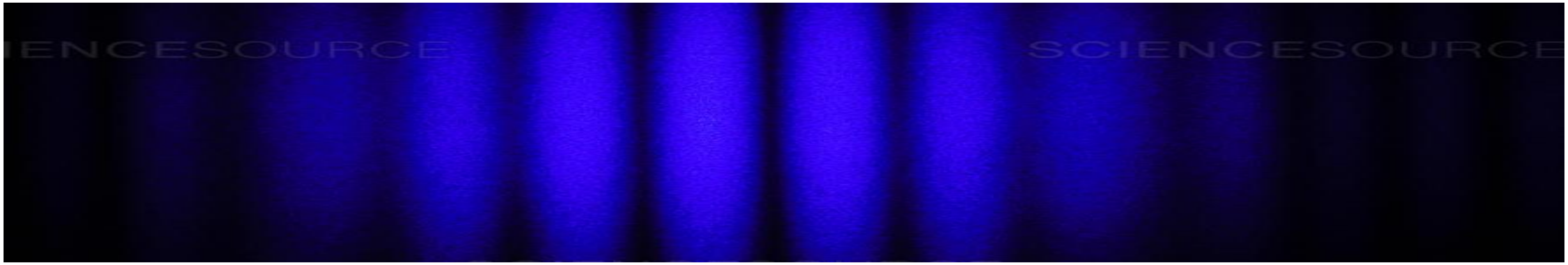


Poynting vector (power density):

$$\langle \boldsymbol{S} \rangle = \left\langle \frac{1}{\mu_0} \boldsymbol{E} \times \boldsymbol{B} \right\rangle = \frac{1}{\mu_0} |E_0 B_0| \langle \sin^2 \omega t \rangle = \frac{1}{2} \frac{1}{c\mu_0} (E_0)^2 = \frac{1}{2} \varepsilon_0 c (E_0)^2$$

4.2 SUPERPOSITION

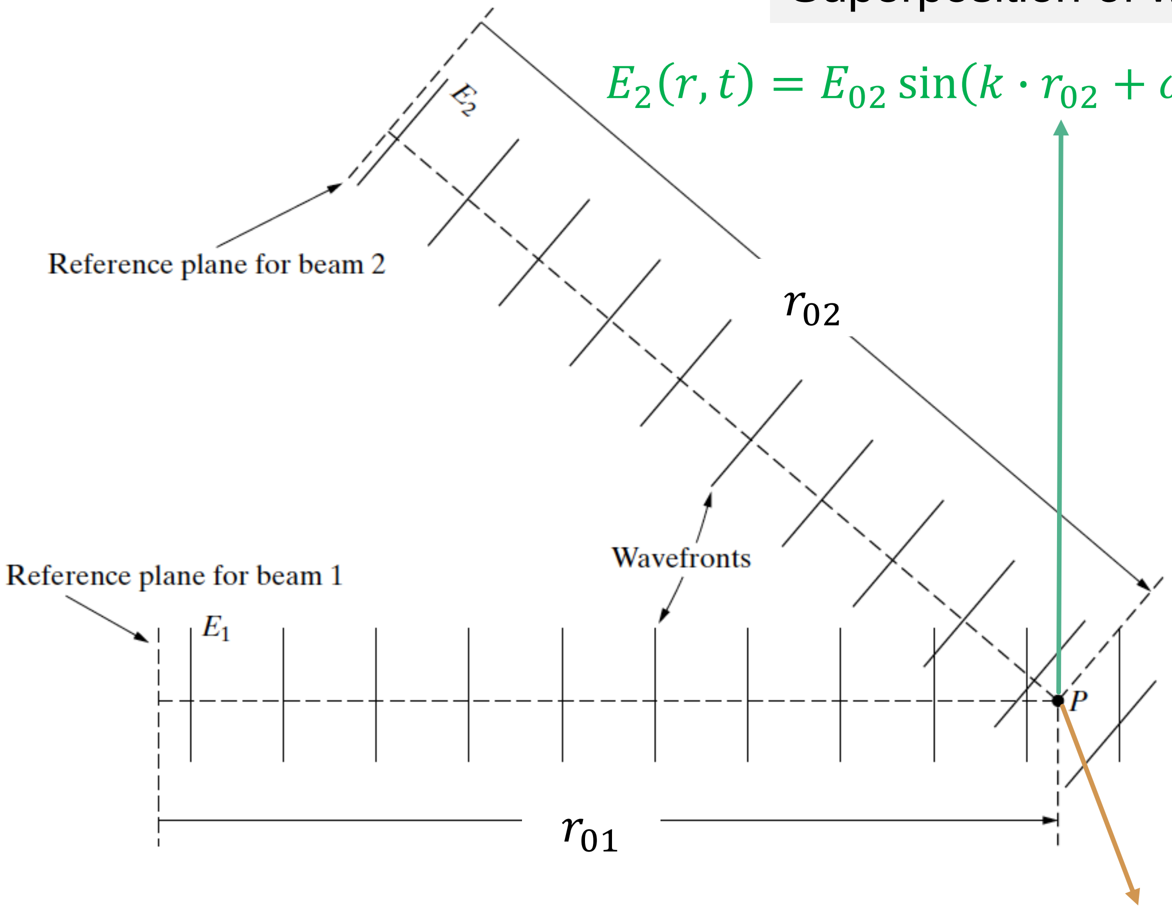
EM waves



4.2 SUPERPOSITION

Superposition of waves of the same frequency

$$E_2(r,t) = E_{02}\sin(k\cdot r_{02} + \omega t + \varphi_{02}) = E_{02}\sin(\omega t + \alpha_{02})$$



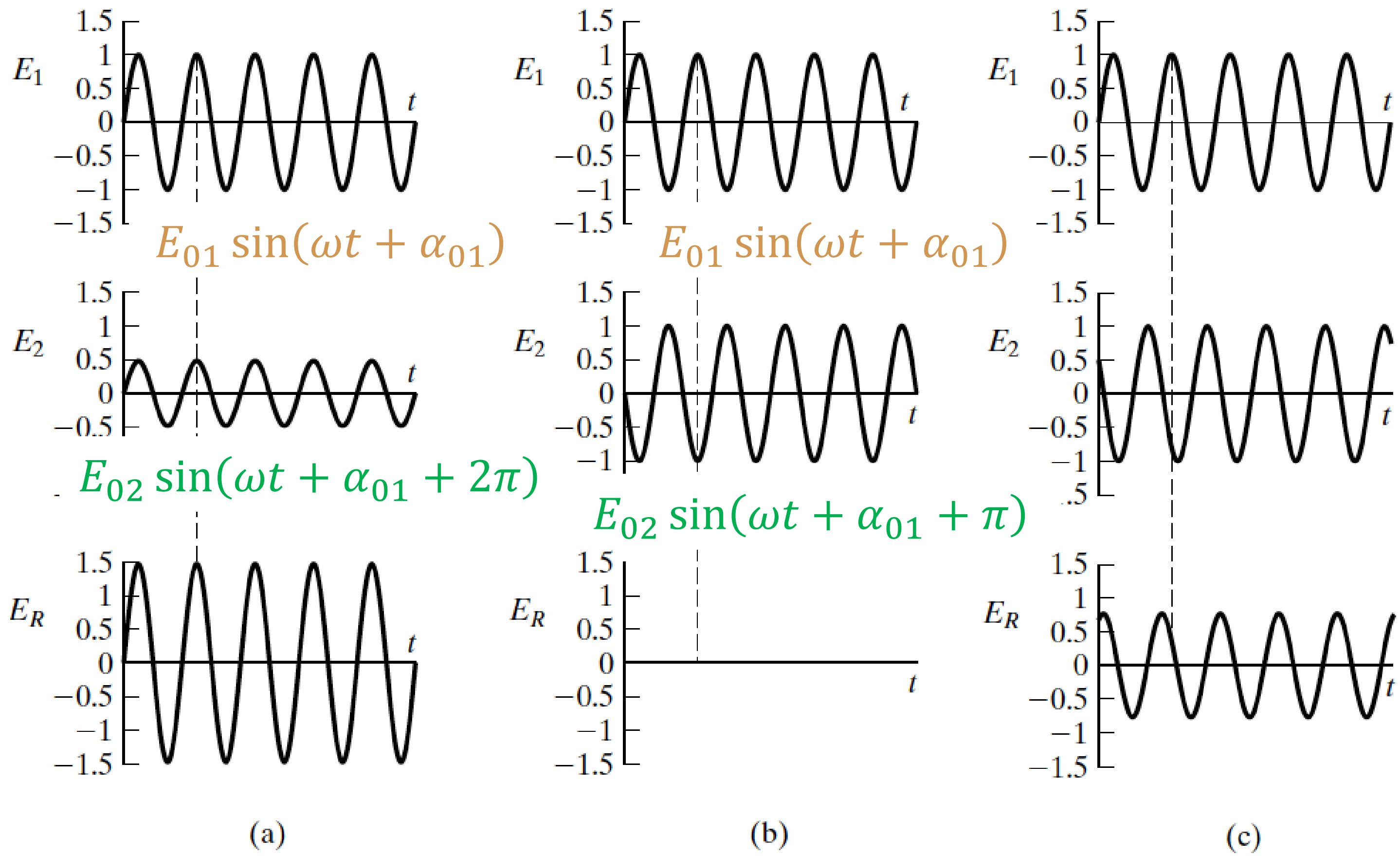
$$E_1(r,t) = E_{01}\sin(k\cdot r_{01} + \omega t + \varphi_{01}) = E_{01}\sin(\omega t + \alpha_{01})$$

4.2 SUPERPOSITION

Constructive

Destructive

General superposition



$$E_{01}\sin(\omega t+\alpha_{01})$$

$$E_{02}\sin(\omega t+\alpha_{01}+2\pi)$$

$$E_{01}\sin(\omega t+\alpha_{01})$$

$$E_{02}\sin(\omega t+\alpha_{01}+\pi)$$

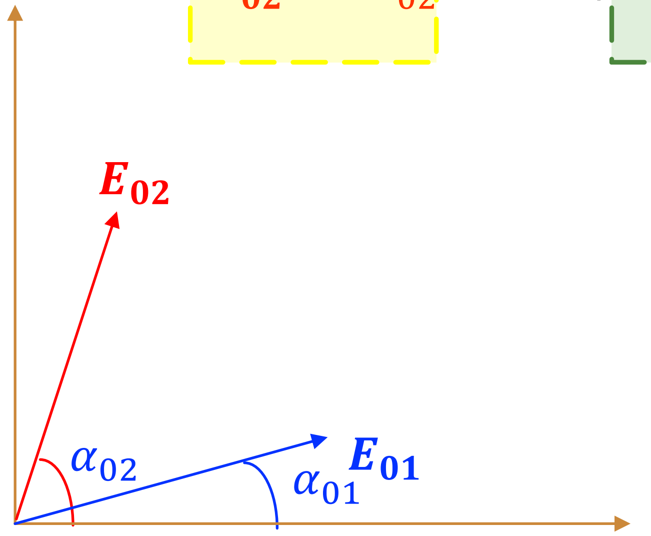
4.2 SUPERPOSITION

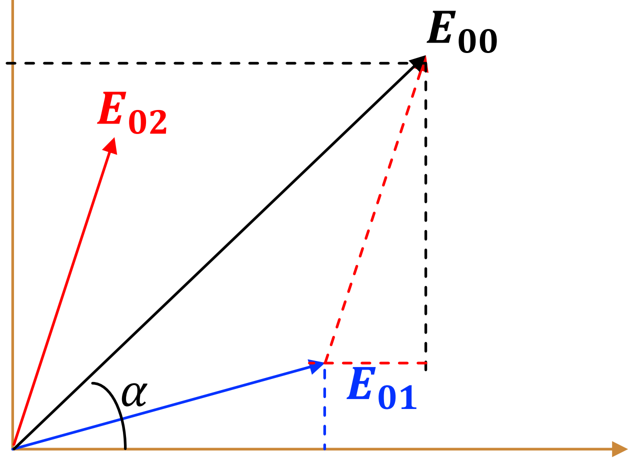
Superposition of waves of the same frequency

$$\boldsymbol{E_1}(\boldsymbol{r},\boldsymbol{t}) = \boldsymbol{E_{01}}\sin(k\cdot r_{01}+\omega t+\varphi_{01}) = \boldsymbol{E_{01}}\sin(\omega t+\alpha_{01})$$
$$= \boxed{\boldsymbol{E_{01}}\cos\alpha_{01}}\sin\omega t + \boxed{\boldsymbol{E_{01}}\sin\alpha_{01}}\cos\omega t$$

$$+$$

$$\boldsymbol{E_2}(\boldsymbol{r},\boldsymbol{t}) = \boldsymbol{E_{02}}\sin(k\cdot r_{02}+\omega t+\varphi_{02}) = \boldsymbol{E_{02}}\sin(\omega t+\alpha_{02})$$
$$= \boxed{\boldsymbol{E_{02}}\cos\alpha_{02}}\sin\omega t + \boxed{\boldsymbol{E_{02}}\sin\alpha_{02}}\cos\omega t$$





4.2 SUPERPOSITION

Superposition of waves of the same frequency

$$\boldsymbol{E_r}(\boldsymbol{r},t)=\boldsymbol{E_1}(\boldsymbol{r},t)+\boldsymbol{E_2}(\boldsymbol{r},t)$$

$$=(\boldsymbol{E_{01}}\cos\alpha_{01}+\boldsymbol{E_{02}}\cos\alpha_{02})\sin\omega t+(\boldsymbol{E_{01}}\sin\alpha_{01}+\boldsymbol{E_{02}}\sin\alpha_{02})\cos\omega t$$

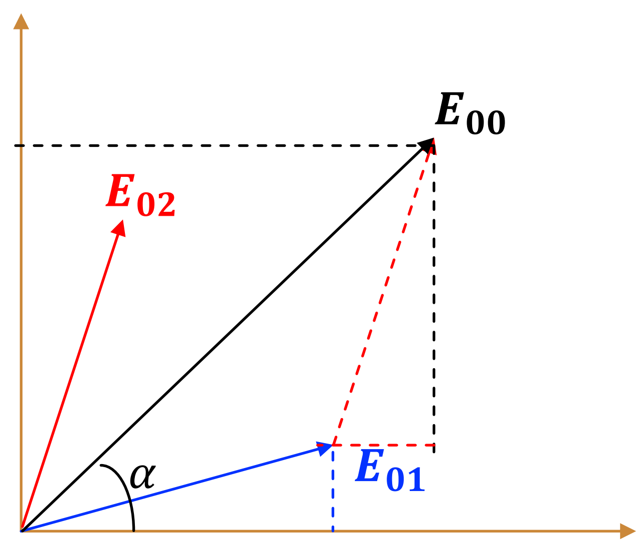
$$=\boldsymbol{E_{00}}\cos\alpha\sin\omega t+\boldsymbol{E_{00}}\sin\alpha\cos\omega t$$

$$=\boldsymbol{E_{00}}\sin(\omega t+\alpha)$$

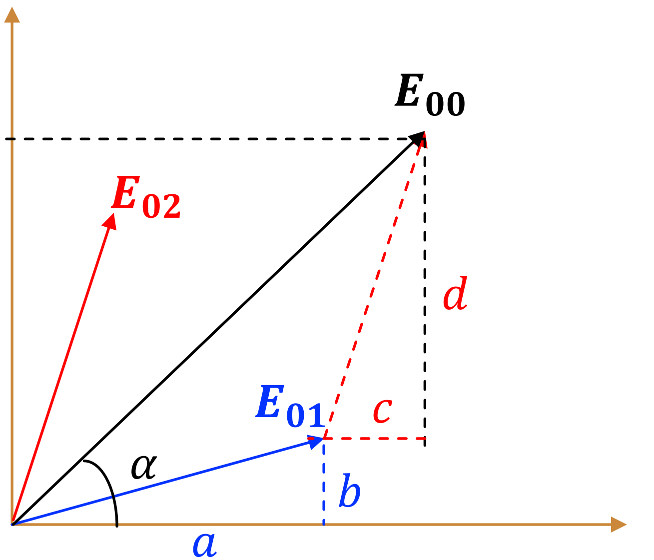
$$\tan\alpha=\frac{b+d}{a+c}=\frac{E_{01}\sin\alpha_{01}+E_{02}\sin\alpha_{02}}{E_{01}\cos\alpha_{01}+E_{02}\cos\alpha_{02}}$$

$$\boldsymbol{E_1}(\boldsymbol{r},t)=\overset{a}{\boldsymbol{E_{01}}\cos\alpha_{01}}\sin\omega t+\overset{b}{\boldsymbol{E_{01}}\sin\alpha_{01}}\cos\omega t$$

$$\boldsymbol{E_2}(\boldsymbol{r},t)=\overset{c}{\boldsymbol{E_{02}}\cos\alpha_{02}}\sin\omega t+\overset{d}{\boldsymbol{E_{02}}\sin\alpha_{02}}\cos\omega t$$



4.2 SUPERPOSITION



$$I \propto {E_{00}}^2$$

$$ {E_{00}}^2 = (a+c)^2 + (b+d)^2$$

$$= a^2 + b^2 + c^2 + d^2 + 2(\mathrm{ac} + \mathrm{bd})$$

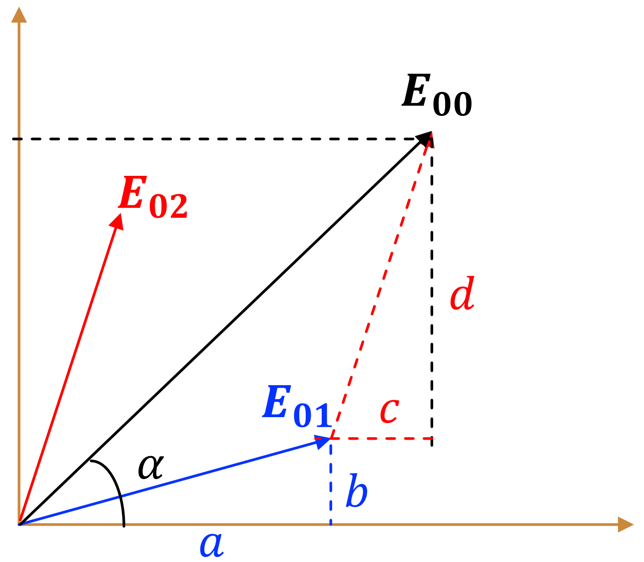
$$= {E_{01}}^2 + {E_{02}}^2 + 2E_{01}E_{02}(\cos\alpha_{01}\cos\alpha_{02} + \sin\alpha_{01}\sin\alpha_{02})$$

$$= {E_{01}}^2 + {E_{02}}^2 + 2E_{01}E_{02}\cos(\alpha_{01} - \alpha_{02})$$

4.2 SUPERPOSITION

$$\tan\alpha = \frac{b+d}{a+c} = \frac{E_{01}\sin\alpha_{01} + E_{02}\sin\alpha_{02}}{E_{01}\cos\alpha_{01} + E_{02}\cos\alpha_{01}}$$

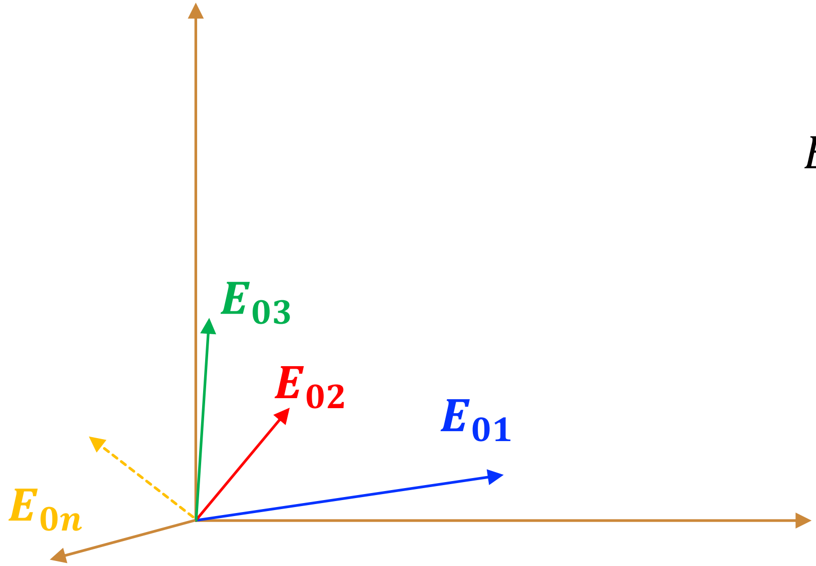
$$I \propto {E_{00}}^2$$

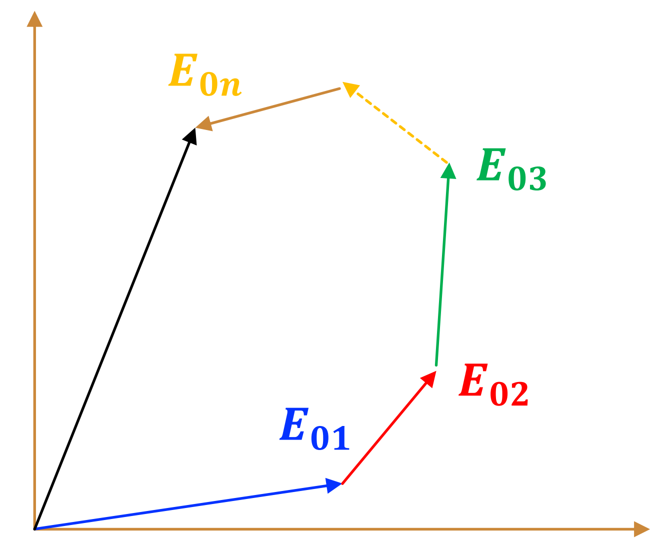


$$\begin{aligned} {E_{00}}^2 &= (a+c)^2 + (b+d)^2 \\ &= (E_{01}\sin\alpha_{01} + E_{02}\sin\alpha_{02})^2 + (E_{01}\cos\alpha_{01} + E_{02}\cos\alpha_{01})^2 \end{aligned}$$

4.2 SUPERPOSITION

$$E_{00}{}^{2} = \left(\sum_{i=1}^{n} E_{0i} \sin \alpha_{0i}\right)^{2} + \left(\sum_{i=1}^{n} E_{0i} \cos \alpha_{0i}\right)^{2}$$





4.2 SUPERPOSITION

$$E_{00}{}^2 = \left(\sum_{i=1}^{n} E_{0i} \sin \alpha_{0i}\right)^2 + \left(\sum_{i=1}^{n} E_{0i} \cos \alpha_{0i}\right)^2$$

$$\left(\sum_{i=1}^{n} E_{0i} \sin \alpha_{0i}\right)^2 = \sum_{i}^{n} E_{0i}{}^2 (\sin \alpha_{0i})^2 + 2 \sum_{i>j}^{n} \sum_{j=1}^{n} E_{0i} E_{0j} \sin \alpha_{0i} \sin \alpha_{0j}$$

$$\left(\sum_{i=1}^{n} E_{0i} \cos \alpha_{0i}\right)^2 = \sum_{i}^{n} E_{0i}{}^2 (\cos \alpha_{0i})^2 + 2 \sum_{i>j}^{n} \sum_{j=1}^{n} E_{0i} E_{0j} \cos \alpha_{0i} \cos \alpha_{0j}$$

$$E_{00}{}^2 = \sum_{i}^{n} E_{0i}{}^2 + 2 \sum_{i>j}^{n} \sum_{j=1}^{n} E_{0i} E_{0j} \left(\sin \alpha_{0i} \sin \alpha_{0j} + \cos \alpha_{0i} \cos \alpha_{0j}\right)$$

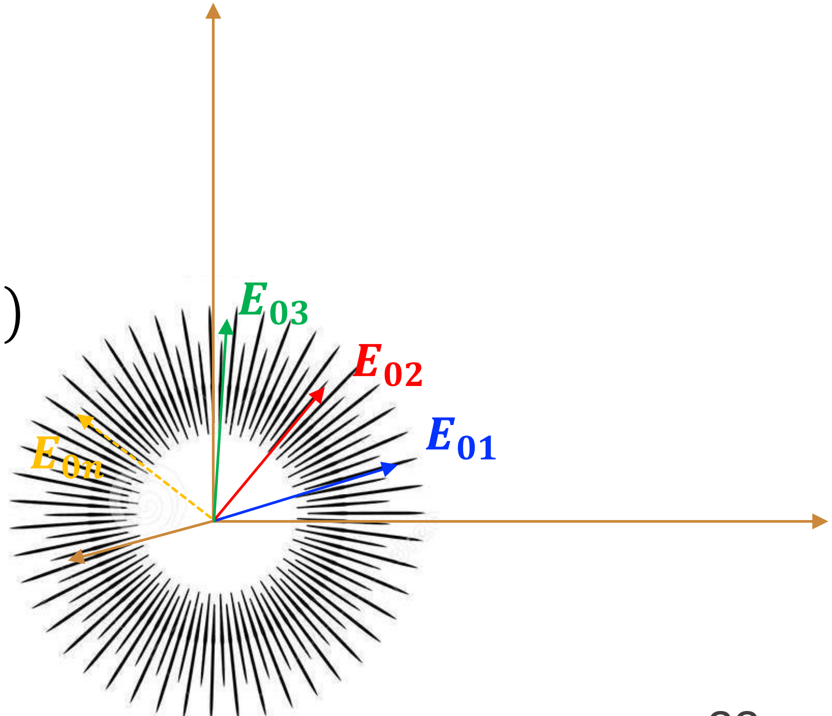
4.2 SUPERPOSITION

$$E_{00}{}^2 = \sum_{i}^{n} E_{0i}{}^2 + 2\sum_{i>j}^{n}\sum_{j=1}^{n} E_{0i}E_{0j}\cos(\alpha_{0i} - \alpha_{0j})$$

Random phased light with equal amplitude

$$E_{00}{}^2 = \sum_{i}^{n} E_0{}^2 + 2\sum_{i>j}^{n}\sum_{j=1,}^{n} E_0{}^2\cos(\alpha_{0i} - \alpha_{0j})$$

$$E_{00}{}^2 = \sum_{i}^{n} E_0{}^2 + 0 = nE_0{}^2$$



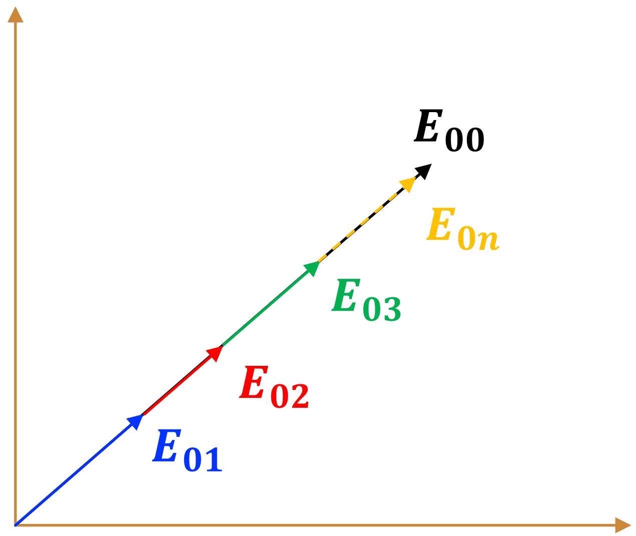
4.2 SUPERPOSITION

Coherent light

$$\boldsymbol{E_1}(\boldsymbol{r},t) = \boldsymbol{E_{01}} \sin(\boldsymbol{k}\cdot\boldsymbol{r_{01}} + \omega t + \varphi_{01}) = \boldsymbol{E_{01}} \sin(\omega t + \alpha_0)$$

$$\vdots$$

$$\boldsymbol{E_i}(\boldsymbol{r},t) = \boldsymbol{E_{0i}} \sin(\boldsymbol{k}\cdot\boldsymbol{r_{0i}} + \omega t + \varphi_{0i}) = \boldsymbol{E_{0i}} \sin(\omega t + \alpha_0)$$

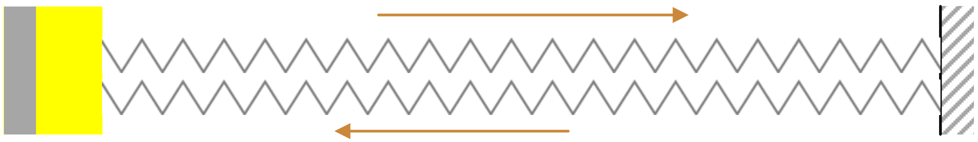


$${E_{00}}^2 = \left(\sum_i^{\infty} E_{0i}\right)^2$$

Coherent light with equal amplitude

$${E_{00}}^2 = \left(\sum_i^{\infty} E_0\right)^2 = N^2 {E_0}^2$$

4.2 SUPERPOSITION-S.W



Forward: $\boldsymbol{E_1}(\boldsymbol{r}, t) = \boldsymbol{E_0} \sin(kx - \omega t)$

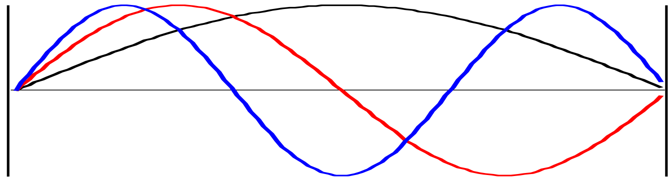
backward: $\boldsymbol{E_2}(\boldsymbol{r}, t) = \boldsymbol{E_0} \sin(kx + \omega t)$

$$\boldsymbol{E_r}(\boldsymbol{r}, t) = \boldsymbol{E_1} + \boldsymbol{E_2} = \boldsymbol{2E_0} \sin\left(\frac{kx - \omega t + kx + \omega t}{2}\right) \cos\left(\frac{kx - \omega t - kx - \omega t}{2}\right)$$

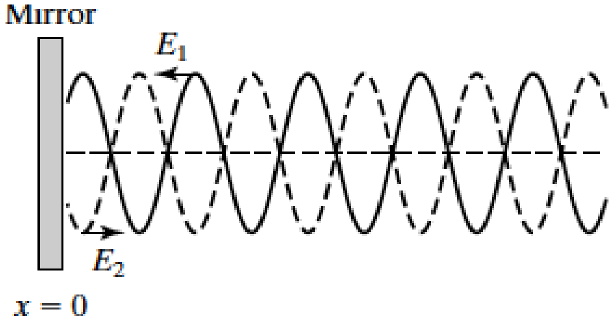
$$\boldsymbol{E_r}(\boldsymbol{r}, t) = \boldsymbol{E_1} + \boldsymbol{E_2} = \boldsymbol{2E_0} \sin(kx) \cos(\omega t)$$

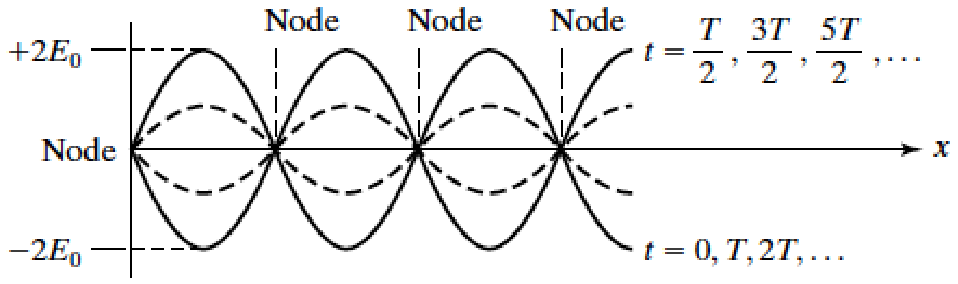
4.2 SUPERPOSITION-S.W

$$u_m(x,t) = c_m \sin(k_m x)\cos(\omega_m t)$$



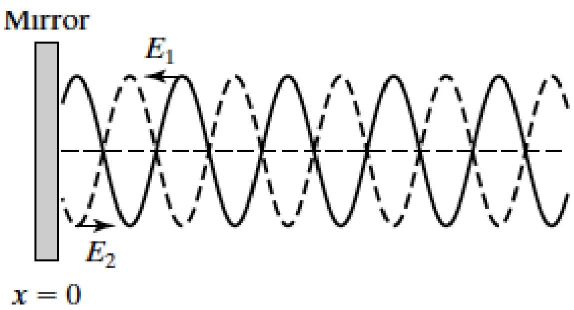
Standing waves $$\boldsymbol{E_r}(\boldsymbol{r},t) = \boldsymbol{E_1} + \boldsymbol{E_2} = \boldsymbol{2E_0} \sin(kx) \cos(\omega t)$$

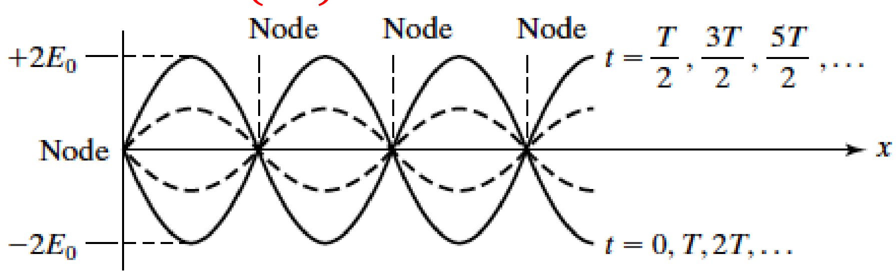




4.2 SUPERPOSITION-S.W

$\sin(kx) = 0$





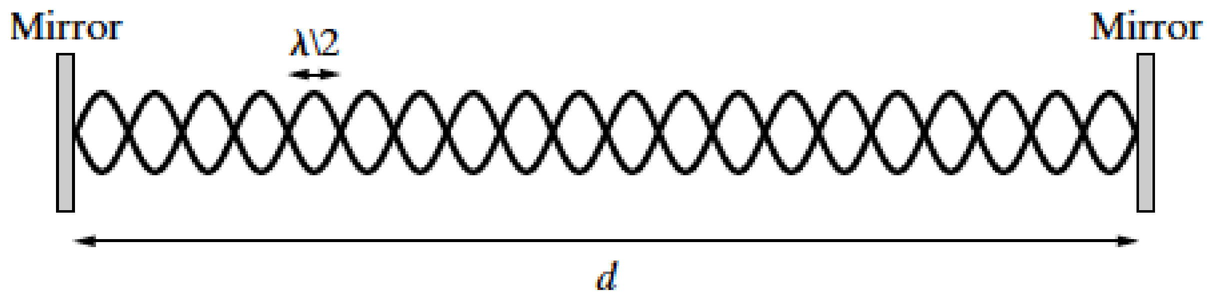
Standing waves $\boldsymbol{E_r}(\boldsymbol{r}, t) = \boldsymbol{E_1} + \boldsymbol{E_2} = \boldsymbol{2E_0} \sin(kx) \cos(\omega t)$

Nodes $kx = m\pi$ $x = m\frac{\pi}{k} = m\frac{\lambda}{2} = 0, \frac{\lambda}{2}, \lambda \ldots$

Maxima $\cos(\omega t) = 1$ $t = m\frac{T}{2} = 0, \frac{T}{2}, T \ldots$

4.2 SUPERPOSITION-S.W

Laser Cavity



The requirement that nodes exist at the mirror positions restricts the wavelengths that can be supported by the cavity to discrete values. If the distance between the cavity mirrors is *d*, the cavity will support standing waves with wavelengths that satisfy the relation

$$d = m\frac{\lambda}{2} \qquad \lambda_m = \frac{2d}{m}$$

The frequencies of the standing wave modes of a laser cavity

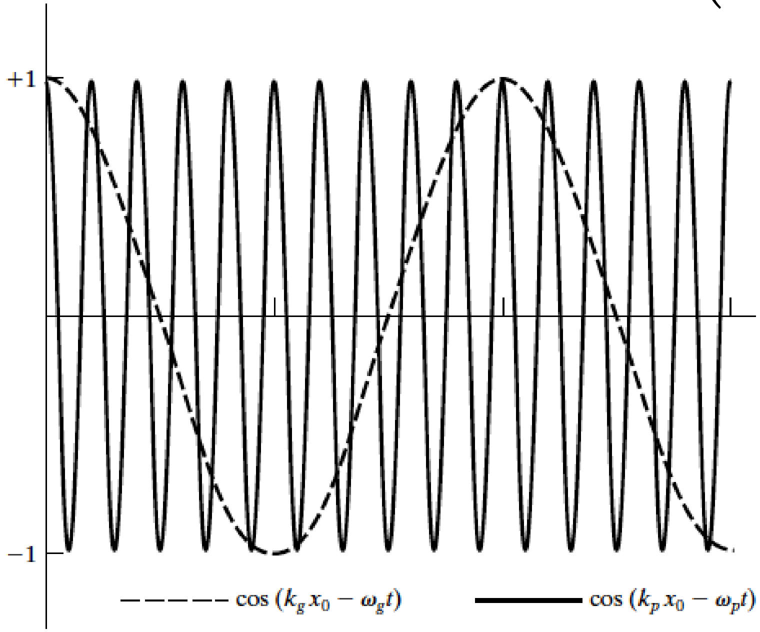
$$\nu_m = \frac{c}{\lambda_m} = m\frac{c}{2d}$$

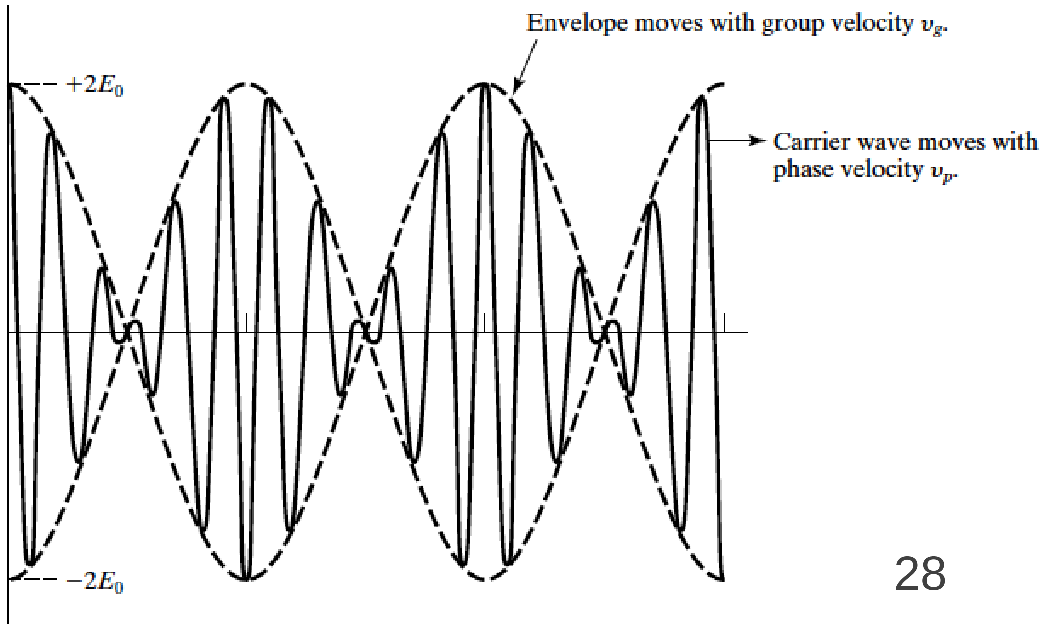
4.2 SUPERPOSITION-BEATING

Beating

$$\boldsymbol{E_1}(\boldsymbol{r},t) = \boldsymbol{E_0}\cos(k_1 x - \omega_1 t) \qquad \boldsymbol{E_2}(\boldsymbol{r},t) = \boldsymbol{E_0}\cos(k_2 x - \omega_2 t)$$

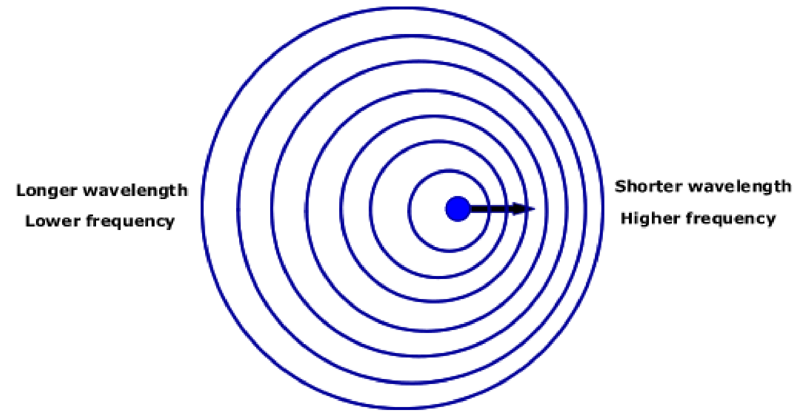
$$\boldsymbol{E_r}(\boldsymbol{r},t) = \boldsymbol{E_1} + \boldsymbol{E_2} = \boldsymbol{2E_0}\cos\left(\frac{k_1+k_2}{2}x - \frac{\omega_1+\omega_2}{2}t\right)\cos\left(\frac{k_1-k_2}{2}x - \frac{\omega_1-\omega_2}{2}t\right)$$

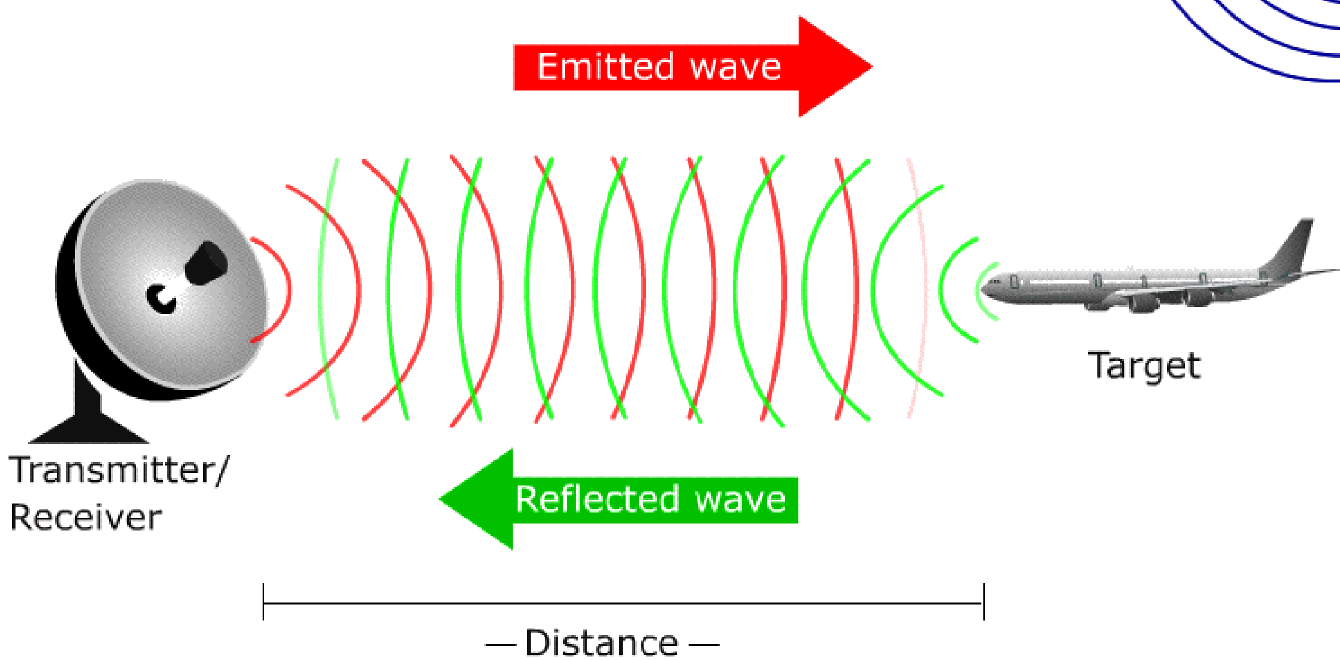




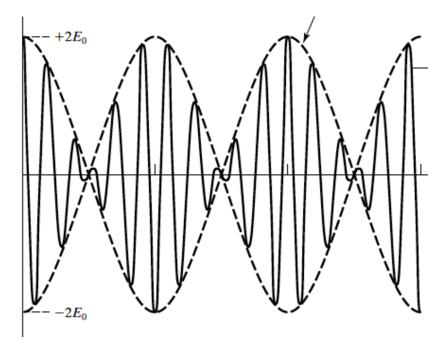
4.2 SUPERPOSITION-BEATING

Doppler radar





$$\omega_1 \sim \omega_2 = \omega$$



4.2 SUPERPOSITION-GV

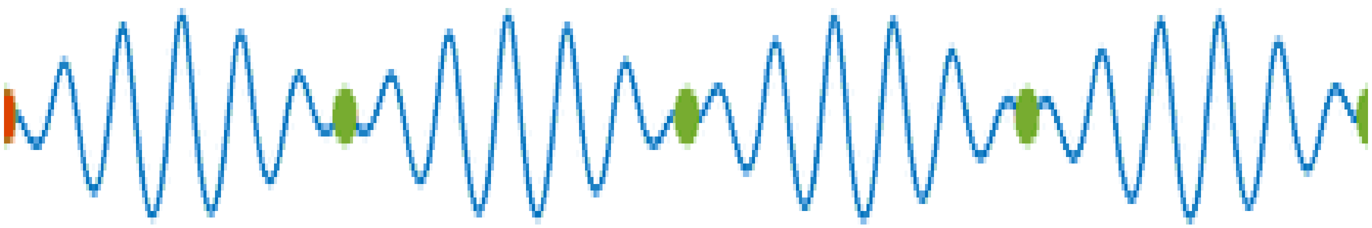
Phase and group velocity

$$\boldsymbol{E_r}(\boldsymbol{r}, t) = \boldsymbol{2E_0} \cos\left(\frac{k_1 - k_2}{2}x - \frac{\omega_1 - \omega_2}{2}t\right) \cos\left(\frac{k_1 + k_2}{2}x - \frac{\omega_1 + \omega_2}{2}t\right)$$

$$= \boldsymbol{2E_0} \cos(k_g t - \omega_g t) \cos(k_p t - \omega_p t)$$

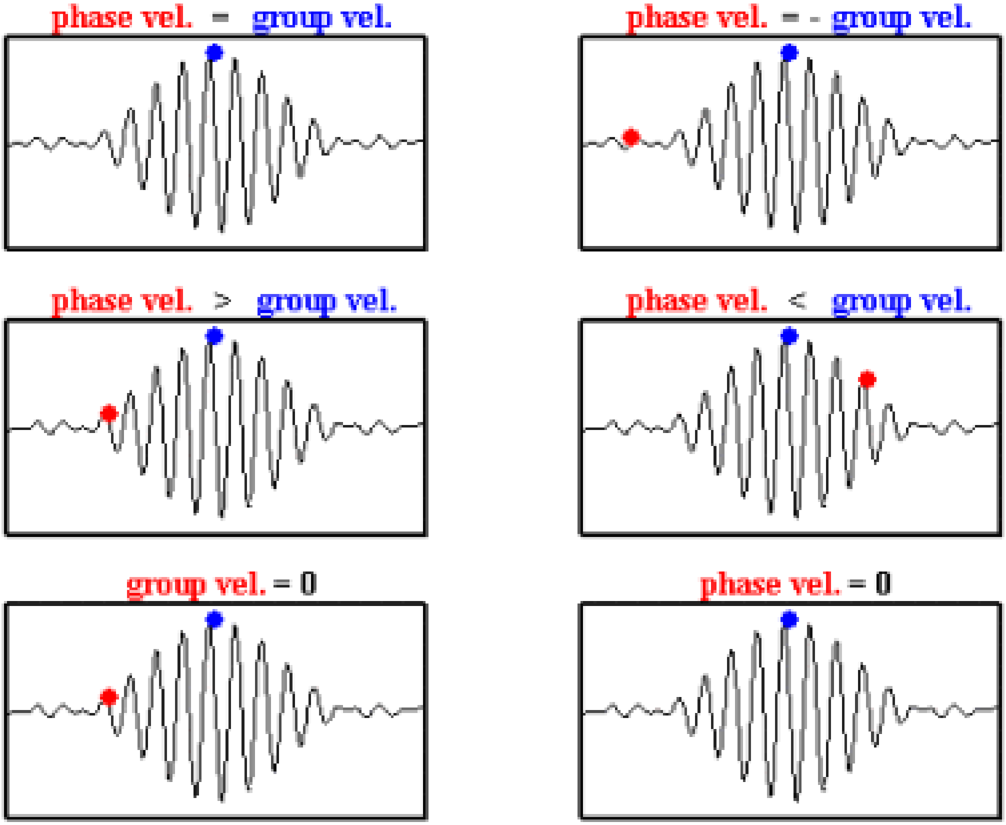
$$v_g = \frac{\omega_g}{k_g} \qquad v_p = \frac{\omega_p}{k_p}$$

Envolope function Carrier function



4.2 SUPERPOSITION-GV

Phase and group velocity



4.2 SUPERPOSITION-GV

Phase and group velocity

$$\boldsymbol{E_r}(\boldsymbol{r},t) = \boldsymbol{2E_0}\cos\left(\frac{k_1-k_2}{2}x - \frac{\omega_1-\omega_2}{2}t\right)\cos\left(\frac{k_1+k_2}{2}x - \frac{\omega_1+\omega_2}{2}t\right)$$

$$= \boldsymbol{2E_0}\cos(k_g t - \omega_g t)\cos(k_p t - \omega_p t)$$

$\omega_1 \sim \omega_2 = \omega$

$$v_g = \frac{\omega_g}{k_g} = \frac{\omega_1-\omega_2}{k_1-k_2} \approx \frac{d\omega}{dk} \qquad v_p = \frac{\omega}{k} = \frac{c}{n} \qquad d\omega = d(kv_p) = v_p dk + k dv_p$$

$$v_g = v_p + k\frac{dv_p}{dk} = v_p - \frac{kc}{n^2}\frac{dn}{dk} = v_p\left(1 - \frac{k}{n}\frac{dn}{dk}\right)$$

$$v_g = v_p\left(1 + \frac{\lambda}{n}\frac{dn}{d\lambda}\right)$$

4.2 SUPERPOSITION-GV

Chromatic **dispersion**

$$\boldsymbol{E_r}(\boldsymbol{r},t) = \boldsymbol{2E_0}\cos\left(\frac{k_1-k_2}{2}x-\frac{\omega_1-\omega_2}{2}t\right)\cos\left(\frac{k_1+k_2}{2}x-\frac{\omega_1+\omega_2}{2}t\right)$$

$$= \boldsymbol{2E_0}\cos(k_g t-\omega_g t)\cos(k_p t-\omega_p t)$$

$$\omega_1 \sim \omega_2 = \omega$$

$$v_g = v_p\left(1+\frac{\lambda}{n}\frac{dn}{d\lambda}\right)$$

Normal dispersion $\frac{dn}{d\lambda} < 0$

$$v_g < v_p$$

Anomalous dispersion $\frac{dn}{d\lambda} > 0$

$$v_g > v_p$$